Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/27/2017
PI Name:	Poirier, David Sc.D.		
Project Title:	Directional Solidification Experiments on the International Space Station		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	07/18/2014	**End Date:**	09/30/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Grugel, Richard	**Contact Phone:**	256 544-9165
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2016 per NSSC (Ed., 7/5/16)		
Key Personnel Changes/Previous PI:	May 2016 report: Dr. N. Swintek who was co-Principal Investigator (PI) resigned his position at the university. The new co-PI is Dr. V. Manga, who is Assistant Research Professor.		
COI Name (Institution):	Tewari, Surendra (Cleveland State University) Manga, Venkat Ph.D. (The University of Arizona)		
Grant/Contract No.:	NNX14AN87G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 8/25/2014): This project is a continuation of "Research for ESA Space Experiments on Directional Solidification of Alloys," grant NNX08AN49G, with period of performance of 5/15/2008-06/30/2014. The end date for the extended grant (i.e., NNX14AN87G) was 09/30/2016. We are studying directional solidification (DS) of alloys solidified in presence of 1-g (on Earth) and in 0-g (in space on the International Space Station-ISS). In this way we can compare the microstructures and macrostructures of samples solidified on Earth and in space to see the effects of convection. We have been carrying out the study by examining the DS-alloy and by computer simulations. During the period of the extension, important properties needed for assessing the microstructures and macrostructures have been studied. Based on the research, it is clear that thermosolutal convection, which always occurs when alloys solidify on Earth, masks the effects of diffusional transport during solidification. With terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, directional solidification occurs with almost no thermosolutal convection, so the convection and attendant macrosegregation are almost completely eliminated, and the growth of the cast dendritic-microstructure is controlled by a diffusional processes. We are part of MICAST (Microstructure Formation in Casting of Technical Alloys) led by European Space Agency (ESA)-researchers in the German Aerospace Center in Germany.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is relevant to a process known as directional solidification, which is used for producing critical castings that are used in gas turbine engines. Gas turbine engines power aircraft (i.e., often called "jet engines") and are also used to generate electric power. These castings are made of vacuum melted alloys that are capable of withstanding the demanding conditions at high temperatures and consequently the metallurgical structure of the alloys must be of the highest quality. The research is leading to an improved understanding of how microstructures and segregation form in directionally solidified castings; hence the science of producing such castings will be enhanced as a result of the research. In addition, the research findings relate to many other types of castings, including billets and ingots, of alloys (i.e., steel, aluminum alloys, etc.) that are important in the "manufacturing chain."
Task Progress:	JANUARY 2017--SUPPLEMENTAL REPORTING FOR FINAL REPORT We studied three important alloy properties for the Al-7%Si alloy that are needed to assess the results of samples that have been returned to Earth after solidification aboard the International Space Station. The properties are the diffusion coefficient of Si in the liquid alloy, the energy of the solid-liquid interface, and the surface tension of the liquid. The first two relate directly to the microstructure that develops during directional solidification, and the third property has been used to evaluate whether separation of the alloy from its crucible-mold has affected the results obtained in the space-grown samples. MAY 2016 ANNUAL REPORTING We have done a detailed microstructure analysis of directionally solidified (DS) Al-7Si alloys processed in microgravity aboard the International Space Station (ISS) and similar ground based experiments at Cleveland State University (CSU). NASA and ESA have collaborated for three DS experiments with Al- 7 wt. % Si alloy aboard the International Space Station (ISS); these are MICAST6, MICAST7, and MICAST12. The first two experiments were processed on the ISS in 2009 and 2010. MICAST12 was processed aboard the ISS in the spring of 2014, but the analyses of the microstructures are still in progress. Major goals are to understand the effect of convection in primary dendrite arm spacings (PDAS) and radial macrosegregation, which are two important features of the directionally solidified castings. Samples taken from the MICAST6 and MICAST7 experiments were used to measure the PDAS and radial macrosegregation and compared to the results obtained for the castings solidified on Earth at CSU. To represent PDAS in DS alloy in the presence of no convection for solidification in space, a theoretical model was used to predict the PDAS. During steady state in the microgravity environment on the ISS, there was a reasonable agreement between the measured and calculated PDAS. There was separation between the sample and the crucible-mold and some convection, however, which probably contributed to altering the PDAS. In ground-based experiments, transverse sections exhibited obvious radial macrosegregation caused by thermosolutal convection resulting in non-agreement with the theoretical model. Using a combination of image processing techniques and electron microprobe analyses, the extent of radial macrosegregation was found to be a function of processing conditions and PDAS.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Ghods M, Johnson L, Lauer M, Grugel RN, Tewari SN, Poirier DR. "Radial macrosegregation and dendrite clustering in directionally solidified Al–7Si and Al–19Cu alloys." Journal of Crystal Growth. 2016 May;441:107-16. http://dx.doi.org/10.1016/j.jcrysgro.2016.02.014 , May-2016
Dissertations and Theses	Angart SG. "Microstructure analysis of aluminium alloy directionally solidified abroad the International Space Station." Masters Thesis, The University of Arizona, Tucson, Arizona, 2015. , Jun-2015